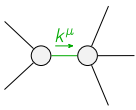
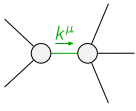


$\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^-}^{\#1} \quad \mathcal{K}_{0^-}^{\#1} \quad \alpha\beta\chi$

$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1} \alpha\beta$	$\mathcal{K}_{1^+}^{\#2} \alpha\beta$	$\mathcal{K}_{1^+}^{\#1} \alpha$	$\mathcal{K}_{1^+}^{\#2} \alpha$
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_1 k^2}{2}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	$\frac{\Upsilon_2 k^2}{2}$	0	0		
$\mathcal{K}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-k^2 \frac{{}^{(4)}\kappa_2}{2}$	$\frac{k^2}{\sqrt{2}} \frac{{}^{(4)}\kappa_2}{2}$		
$\mathcal{K}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{k^2}{\sqrt{2}} \frac{{}^{(4)}\kappa_2}{2}$	$-\frac{k^2}{2} \frac{{}^{(4)}\kappa_2}{2}$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta\chi$
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{3\Upsilon_3 k^2}{2}$	0	
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

$\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^-}^{\#1} \quad \mathcal{J}_{0^-}^{\#1} \quad \alpha\beta\chi$

$\mathcal{J}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1} \alpha\beta$	$\mathcal{J}_{1^+}^{\#2} \alpha\beta$	$\mathcal{J}_{1^+}^{\#1} \alpha$	$\mathcal{J}_{1^+}^{\#2} \alpha$
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2} \text{Det}(1^+)}$	$-\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-\frac{4}{9 k^2} \frac{{}^{(4)}\kappa_2}{2}$	$\frac{2}{9 k^2} \frac{{}^{(4)}\sqrt{2}}{\kappa_2}$		
$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{2}{9 k^2} \frac{{}^{(4)}\sqrt{2}}{\kappa_2}$	$-\frac{2}{9 k^2} \frac{{}^{(4)}\kappa_2}{\kappa_2}$	$\mathcal{J}_{2^+}^{\#1} \alpha\beta$	$\mathcal{J}_{2^+}^{\#1} \alpha\beta\chi$
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0	
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

Abbreviations used in matrices			
$\Upsilon_1 == 3 \left(\kappa_1 + \kappa_2 \right) + 2 \kappa_3$ && $\Upsilon_2 == \kappa_3 + \kappa_4$ && $\Upsilon_3 == \kappa_1 + \kappa_2$ &&			
$\text{Det}(1^+) == -\frac{1}{8} k^4 \left(6 \kappa_1 \left(\kappa_3 + \kappa_4 \right) + 6 \kappa_2 \left(\kappa_3 + \kappa_4 \right) + \left(2 \kappa_3 + \kappa_4 \right)^2 \right)$ && $\text{Det}(2^+) == \frac{3}{2} k^2 \left(\kappa_1 + \kappa_2 \right)$			
Lagrangian			
$\kappa_1 \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^{\alpha\beta} + \kappa_2 \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^{\alpha\beta} + \kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \kappa_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 3 \kappa_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - 3 \kappa_2 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \frac{1}{2} \kappa_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \kappa_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \quad \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}_{\alpha}^{\alpha\beta} == 0$	
$\mathcal{J}_1^{\#1\alpha} + 2 \mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}_{\beta}^{\beta\chi\alpha} + \partial_\chi \partial^\chi \mathcal{J}_{\beta}^{\beta\alpha\chi} == 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\delta} \epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\alpha}_{\delta} ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta}_{\delta} \epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\beta}_{\delta}$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$(54 \kappa_2^3 + 54 \kappa_2^2 \kappa_3 + 20 \kappa_2 \kappa_3^2 + 36 \kappa_2^2 \kappa_4 + 20 \kappa_2 \kappa_3 \kappa_4 + 5 \kappa_2 \kappa_4^2 + 18 \kappa_1 (3 \kappa_2 + \kappa_3 + \kappa_4) + 3 \kappa_1 (36 \kappa_2^2 + (2 \kappa_3 + \kappa_4)^2 + 6 \kappa_2 (4 \kappa_3 + 3 \kappa_4))) - \sqrt{(324 \kappa_1^4 (9 \kappa_2^2 - 2 \kappa_2 (\kappa_3 + \kappa_4) + (\kappa_3 + \kappa_4)^2) + \kappa_2^2 (2916 \kappa_2^4 + 7 (2 \kappa_3 + \kappa_4)^4 + 648 \kappa_2^3 (5 \kappa_3 + 2 \kappa_4) + 36 \kappa_2 (2 \kappa_3 + \kappa_4)^2 (4 \kappa_3 + 3 \kappa_4) + 108 \kappa_2^2 (15 \kappa_3^2 + 16 \kappa_3 \kappa_4 + 5 \kappa_4^2)) + 9 \kappa_1^2 (1944 \kappa_2^4 + (2 \kappa_3 + \kappa_4)^4 + 216 \kappa_2^3 (4 \kappa_3 + \kappa_4) + 4 \kappa_2 (2 \kappa_3 + \kappa_4)^2 (6 \kappa_3 + 7 \kappa_4) + 24 \kappa_2^2 (7 \kappa_3^2 + 13 \kappa_3 \kappa_4 + 7 \kappa_4^2)) + 108 \kappa_1^3 (108 \kappa_2^2 + 6 \kappa_2 (2 \kappa_3 - \kappa_4) + (\kappa_3 + \kappa_4) (2 \kappa_3 + \kappa_4)^2 + \kappa_2 (4 \kappa_3^2 + 14 \kappa_3 \kappa_4 + 9 \kappa_4^2)) + 12 \kappa_1 \kappa_2 (972 \kappa_2^4 + 21 \kappa_2 (\kappa_3 + \kappa_4) (2 \kappa_3 + \kappa_4)^2 + (2 \kappa_3 + \kappa_4)^4 + 54 \kappa_2^3 (14 \kappa_3 + 5 \kappa_4) + 9 \kappa_2^2 (28 \kappa_3^2 + 34 \kappa_3 \kappa_4 + 13 \kappa_4^2)))/(\kappa_2 (\kappa_1 + \kappa_2) (6 \kappa_1 (\kappa_3 + \kappa_4) + 6 \kappa_2 (\kappa_3 + \kappa_4) + (2 \kappa_3 + \kappa_4)^2))$
	2	0	$(54 \kappa_2^3 + 54 \kappa_2^2 \kappa_3 + 20 \kappa_2 \kappa_3^2 + 36 \kappa_2^2 \kappa_4 + 20 \kappa_2 \kappa_3 \kappa_4 + 5 \kappa_2 \kappa_4^2 + 18 \kappa_1 (3 \kappa_2 + \kappa_3 + \kappa_4) + 3 \kappa_1 (36 \kappa_2^2 + (2 \kappa_3 + \kappa_4)^2 + 6 \kappa_2 (4 \kappa_3 + 3 \kappa_4))) + \sqrt{(324 \kappa_1^4 (9 \kappa_2^2 - 2 \kappa_2 (\kappa_3 + \kappa_4) + (\kappa_3 + \kappa_4)^2) + \kappa_2^2 (2916 \kappa_2^4 + 7 (2 \kappa_3 + \kappa_4)^4 + 648 \kappa_2^3 (5 \kappa_3 + 2 \kappa_4) + 36 \kappa_2 (2 \kappa_3 + \kappa_4)^2 (4 \kappa_3 + 3 \kappa_4) + 108 \kappa_2^2 (15 \kappa_3^2 + 16 \kappa_3 \kappa_4 + 5 \kappa_4^2)) + 9 \kappa_1^2 (1944 \kappa_2^4 + (2 \kappa_3 + \kappa_4)^4 + 216 \kappa_2^3 (4 \kappa_3 + \kappa_4) + 4 \kappa_2 (2 \kappa_3 + \kappa_4)^2 (6 \kappa_3 + 7 \kappa_4) + 24 \kappa_2^2 (7 \kappa_3^2 + 13 \kappa_3 \kappa_4 + 7 \kappa_4^2)) + 108 \kappa_1^3 (108 \kappa_2^2 + 6 \kappa_2 (2 \kappa_3 - \kappa_4) + (\kappa_3 + \kappa_4) (2 \kappa_3 + \kappa_4)^2 + \kappa_2 (4 \kappa_3^2 + 14 \kappa_3 \kappa_4 + 9 \kappa_4^2)) + 12 \kappa_1 \kappa_2 (972 \kappa_2^4 + 21 \kappa_2 (\kappa_3 + \kappa_4) (2 \kappa_3 + \kappa_4)^2 + (2 \kappa_3 + \kappa_4)^4 + 54 \kappa_2^3 (14 \kappa_3 + 5 \kappa_4) + 9 \kappa_2^2 (28 \kappa_3^2 + 34 \kappa_3 \kappa_4 + 13 \kappa_4^2)))/(\kappa_2 (\kappa_1 + \kappa_2) (6 \kappa_1 (\kappa_3 + \kappa_4) + 6 \kappa_2 (\kappa_3 + \kappa_4) + (2 \kappa_3 + \kappa_4)^2))$
Resolved unitarity condition(s):		$(\kappa_4 < 0$ && $((\kappa_3 < -\frac{\kappa_4}{2}$ && $\kappa_2 > 0$ && $-\kappa_2 < \kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4}) \parallel (-\frac{\kappa_4}{2} < \kappa_3 < -\kappa_4$ && $\kappa_2 > 0$ && $-\kappa_2 < \kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4}) \parallel (\kappa_3 == -\kappa_4$ && $\kappa_2 > 0$ && $\kappa_1 > -\kappa_2) \parallel$ $(\kappa_3 > -\kappa_4$ && $\kappa_2 > 0$ && $(\kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4} \parallel (\kappa_1 > -\kappa_2))) \parallel$ $(\kappa_4 == 0$ && $((\kappa_3 < 0$ && $\kappa_2 > 0$ && $-\kappa_2 < \kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2}{6 \kappa_3}) \parallel$ $(\kappa_3 > 0$ && $\kappa_2 > 0$ && $(\kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2}{6 \kappa_3} \parallel (\kappa_1 > -\kappa_2)))) \parallel$ $(\kappa_4 > 0$ && $((\kappa_3 < -\kappa_4$ && $\kappa_2 > 0$ && $-\kappa_2 < \kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4}) \parallel$ $(\kappa_3 == -\kappa_4$ && $\kappa_2 > 0$ && $\kappa_1 > -\kappa_2) \parallel$ $(-\kappa_4 < \kappa_3 < -\frac{\kappa_4}{2}$ && $\kappa_2 > 0$ && $(\kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4} \parallel (\kappa_1 > -\kappa_2))) \parallel$ $(\kappa_3 == -\frac{\kappa_4}{2}$ && $\kappa_2 > 0$ && $(\kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4} \parallel (\kappa_1 > \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4})) \parallel$ $(\kappa_3 > -\frac{\kappa_4}{2}$ && $\kappa_2 > 0$ && $(\kappa_1 < \frac{-6 \kappa_2 \kappa_3 - 4 \kappa_3^2 - 6 \kappa_2 \kappa_4 - 4 \kappa_3 \kappa_4 - \kappa_4^2}{6 \kappa_3 + 6 \kappa_4} \parallel (\kappa_1 > -\kappa_2))))$	